CyberSentry

Learn Cybersecurity step by step

Team Members:-
Gayathri Amith
Sreekuttan B
Amal Mani Joby
Alan Joseph
Sreeraj K R

Problem Statement

Traditional Limitations

Traditional cybersecurity learning platforms often rely on static courses and theoretical content, leaving learners with a limited understanding of practical applications.

Our Solution

Our application introduces a step-by-step task-based learning approach, guiding users from beginner to advanced levels. The platform features an interactive leaderboard to track progress and provides hands-on practice with real-world cybersecurity tools.

Existing System

Online Tutorials

Online tutorials and courses (YouTube, Udemy, Coursera, etc.).

Theoretical Resources

Books and PDFs providing theoretical knowledge.

Hands-on Platforms

Hands-on platforms like TryHackMe and Hack The Box.

Limitations of Existing Systems

Lack of Structure

Lack of structured, step-by-step learning.

High Cost

High cost of some cybersecurity courses.

Limited Simulation

Limited real-world simulation environments.

Beginner Unfriendliness

Not beginner-friendly for those with no prior knowledge.

Proposed System

1 **MERN Stack**

A MERN-stack-based interactive learning platform.

2 **Step-by-Step Learning**

Step-by-step tasks for learning cybersecurity.

3 **Hands-on Exercises**

Hands-on exercises with real-world tools like Nmap.

4 **Progress Tracking**

User progress tracking and certification.

Technology Used

1. **MongoDB**

   MongoDB (Database)

2. **Express.js**

   Express.js (Backend Framework)

3. **React.js**

   React.js (Frontend Framework)

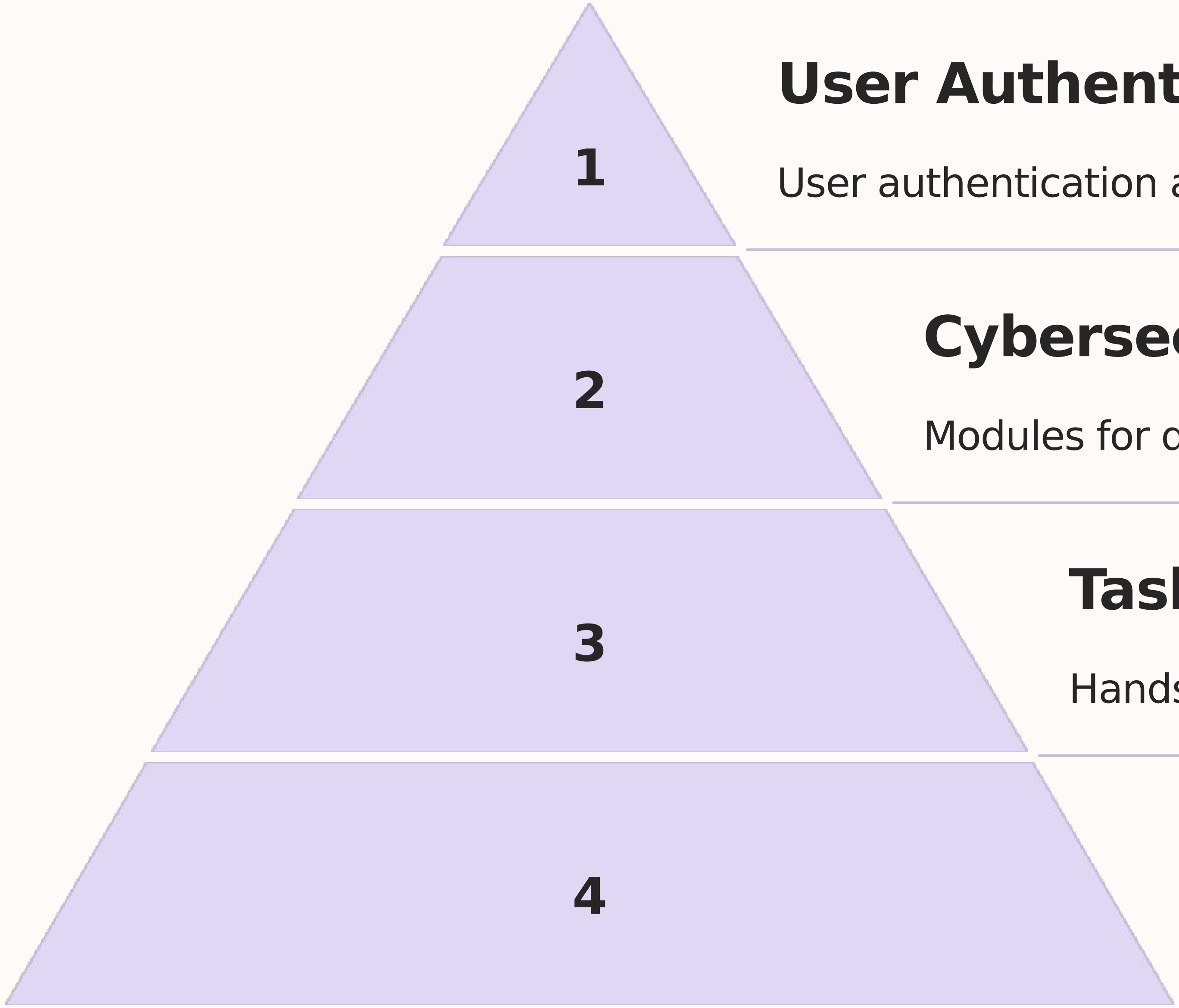
4. **Node.js**

   Node.js (Server-side Runtime)

5. **Tailwind CSS**

   Tailwind CSS (Styling)

Design/ Functionality (Block Diagram)



User Authentication

User authentication and dashboard.

Cybersecurity Modules

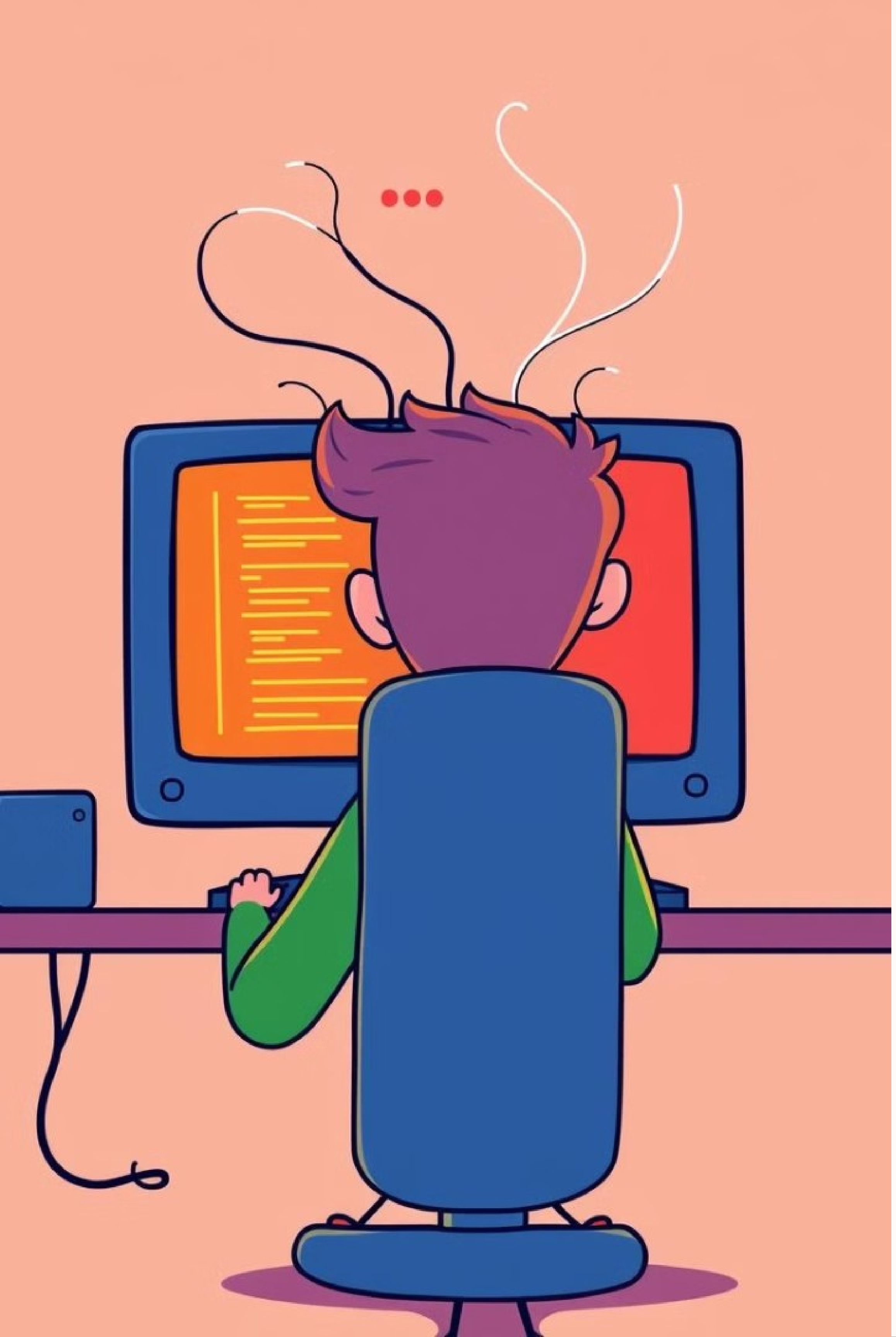
Modules for different cybersecurity topics.

Task Execution

Hands-on task execution and progress tracking.

Database

Database to store user progress.



Final Product

1 **Functional Website**

A fully functional cybersecurity learning website.

2 **Interactive Exercises**

Interactive exercises with real-world cybersecurity tools.

3 **Structured Approach**

A structured approach from beginner to advanced topics.

Conclusion

1

Engaging Learning

Provides an engaging, task-based approach to cybersecurity learning.

2

Hands-on Experience

Helps students and professionals gain hands-on experience.

3

Bridging the Gap

Bridges the gap between theoretical knowledge and practical implementation.

Future Scope

1 AI-based Learning

Adding AI-based personalized learning paths.

2 More Tools

Integrating more cybersecurity tools.

3 Mobile App

Developing a mobile application.

4 Gamification

Gamification elements for better engagement.

References

1

Books and Papers

Books, research papers, and websites used for development.

2

Tools and Technologies

Tools and technologies documentation.

3

Online Platforms

Online cybersecurity learning platforms for reference.

Thank You!